Hari Jatin Dave

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SUMMARY

Results-driven data professional with a Master's in Data Science and experience across analytics, machine learning, and AI solutions. Skilled in Python, SQL, and cloud tools, building scalable data pipelines and models that drive business insights. Proven ability to translate complex data into actionable decisions across healthcare and industrial domains.

TECHNICAL SKILLS

Programming & Data: Python, SQL, R, ETL Pipelines, Data Modeling, EDA, Statistical Analysis
Machine Learning & AI: ML, Predictive Modeling, Time-Series, NLP, Deep Learning (CNN, LSTM), Feature Engineering
Frameworks & Libraries: Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, XGBoost, Flask
Visualization & Tools: Power BI, Tableau, Streamlit, Jupyter, Excel
Cloud & AI Platforms: Azure AI Foundry, LangChain, RAG, OpenAI API, ServiceNow (Virtual Agent, NLU), GCP, Git
Soft Skills: Communication, Problem Solving, Collaboration, Business Acumen

PROFESSIONAL EXPERIENCE

Data/AI Engineer ***August 2025 – Present***
UGenome AI *Tucson, AZ*

- **Designed** scalable **ETL pipelines using Azure Data Factory and SQL** to ingest and integrate structured and unstructured pharmacogenomic datasets, improving data availability and reducing manual preprocessing effort across reporting workflows.
- **Developed** LLM-powered AI agents in **Azure AI Foundry** using retrieval-augmented generation (RAG) to summarize complex genomic reports, enabling more interpretable and structured clinical insights for downstream analysis and decision-making.
- **Implemented** data preprocessing and validation frameworks using Python to standardize clinical data, improving data consistency and ensuring compliance with HIPAA/GDPR requirements for reliable production analytics.
- **Collaborated** with product, engineering, and clinical stakeholders to translate genomic data into deployable AI-driven solutions, improving usability of insights and driving adoption of data-informed workflows across teams.

Data Science Researcher Intern ***August 2024 – January 2025***
University of Arizona *Tucson, AZ*

- **Analyzed** historical mining safety data using **Python and MATLAB (ARIMA/SARIMA)**, identifying a **projected 12% rise in incident rates** and enabling the design of proactive safety protocols and early intervention strategies across operations.
- **Developed** LSTM-based time-series models using **TensorFlow/Keras** to capture temporal injury patterns, enabling earlier risk detection and improving hazard response planning through more accurate scenario forecasting.
- **Engineered** a neural network model in PyTorch to predict SAG mill power consumption from **1,000+ sensor variables**, achieving **~90% predictive accuracy** and supporting more energy-efficient and cost-aware operational optimization.
- **Visualized** safety and operational insights through interactive dashboards and reports, improving stakeholder visibility into key risk indicators and reducing analysis turnaround time by **~30%** for faster decision-making.

Data Science Researcher ***May 2024 – August 2024***
University of Arizona *Tucson, AZ*

- **Engineered** a TensorFlow-based CNN for rock image classification, achieving **85% accuracy** and reducing manual lithology identification effort by **~40% through automated mineral categorization workflows**.
- **Refined** predictive models using feature engineering and statistical techniques (Pearson correlation, mutual information), reducing input dimensionality while maintaining performance and improving overall computational efficiency across experiments.
- **Developed** regression models **(Random Forest, Ridge, SVR)** for mineral resource estimation, achieving **$R^2 \approx 0.92$** and enabling more accurate forecasting to support data-driven planning and operational decision-making.
- **Collaborated** with researchers and domain experts to iterate on model design and validate outputs, improving alignment between ML predictions and real-world geological interpretations for practical deployment in workflows.

Data Science Intern ***December 2022 – June 2023***
Schneider Electric India Pvt. Ltd. *Ahmedabad, India*

- **Engineered** end-to-end pump efficiency models using **Scikit-learn** (Random Forest, XGBoost) and TensorFlow, achieving **96% R^2** and enabling more reliable predictive maintenance planning across critical industrial assets.
- **Designed** an anomaly detection framework using **XGBoost, ADF tests**, and time-series decomposition to identify early fault patterns and support proactive maintenance decisions, reducing risk of unexpected equipment failures.
- **Analyzed** 3M+ industrial sensor records using **Python and SQL** to uncover performance inefficiencies, contributing to **~$500K in projected annual energy savings** through data-driven optimization insights and improved operational strategies.
- **Developed** real-time analytics dashboards using Flask and Plotly to monitor asset performance KPIs, improving visibility for engineering teams and reducing manual reporting effort by **~50% across workflows**.

PROJECTS

MedAssist AI: Context-Aware Medical Assistant Agent ***October 2025***

- Developed a chatbot using Streamlit, LangChain, and Docker to answer medical questions from uploaded PDFs via chat.
- Parsed clinical records with PyMuPDF and applied Ollama + Mistral AI for embedding, summarization, and RAG-based response.
- Integrated the OpenAI API for natural language answers and deployed a secure, containerized backend for local inference.

BookBase AI: MySQL, MongoDB, Streamlit, LangChain ***May 2025***

- Designed comprehensive data models and schemas using MySQL and MongoDB, standardizing data across platforms.
- Built an AI-powered assistant with LangChain + Mixtral-8x7B to convert natural queries into MongoDB pipelines.
- Developed interactive Streamlit dashboards to track sales and customer trends, automating key business insights.

EDUCATION

Master of Science in Data Science ***August 2023 – May 2025***
University of Arizona, Tucson, AZ

Bachelor of Science in Computer Applications ***August 2020 – May 2023***
MS University, Vadodara, India